FSM Controller

FETCH_DECODE / EXECUTE_1 / EXECUTE_2

clk
rst_n
alu_flags
ALUOp
PCSel
ALUASel
ALUBSel[1:0]
MemRead → read_en
AddrSel
RFWrite_n
WBSel
EAWrite
mem_ready
IRWrite
funct7
opcode
funct3
MemWriteMode
PCWrite
3
4
32
next_pc
PC MUX
00: PC+4
01: PC+IMM
10: JALR
pc_plus_4
pc_plus_imm
jalr_address
PC + IMM
instruction_imm
PC + 4
32'd4
pc
Load Alignment
Store Alignment
alu_result
Effective Address Register
EA
2
effective_address_register
PC Register
Address MUX
PC + 4
PC + IMM
ALU A MUX
trap_pc
trap_pc = pc
clk / rst_n shared
Address MUX
0: PC
1: EA
addr[31:0]
load_result
pc_plus_4[31:0]
instruction_u_imm[31:0]
pc_plus_imm[31:0]
alu_result[31:0]
Writeback MUX
000 ALU
001 Load
010 PC+4
011 UIMM
100 PC+IMM
rf_write_data[31:0]
Store Alignment + Byte Write Enable SB SH SW
rf_data_2
write_en
data_out
ALU A MUX
0: rf_data_1
1: pc
alu_in_a
rf_data_1
ALU
alu_flags[2:0]
alu_in_b
ALU B MUX
00: rf_data_2
01: imm
10: 4
rs1[4:0] → read_addr_1
rs2[4:0] → read_addr_2
rd[4:0] → write_addr
Register File
32 × 32
rf_write_data
RFWrite_n
Decoder
instruction_register
instruction_imm
instruction_u_imm
JALR Address
{alu_result[31:2], 2'b00}
Instruction Register
Load Alignment
LB LH LW LBU LHU
data_in
addr
External Memory Interface